Summary in Graph

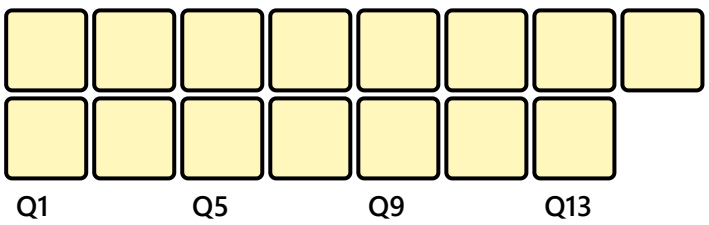
Exam Summary (GO Classes CS/DA | Linear Algebra | Rank, System of equation)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (11 + 4)
Total Marks:	19 (11 + 8)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (42) out of 415 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be an $m \times n$ matrix with real entries. Which of the following statements are true?

A. $\text{rank}(A^T A) \leq \text{rank}(A)$

B. $\text{rank}(A^T A) = \text{rank}(A)$

C. $\text{rank}\left(A^T A\right) > \text{rank}(A)$

D. $\text{rank}\left(A^T A\right) \leq \min(m, n)$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be an $m \times n$ matrix, and let $P \in \mathbb{R}^{m \times m}$ and $Q \in \mathbb{R}^{n \times n}$ be invertible matrices.
Which of the following statements are true?

A. $\text{rank}(PAQ) = \text{rank}(A)$

B. $\text{rank}(PA) = \text{rank}(A)$

C. $\text{rank}(AQ) = \text{rank}(A)$

D. $\text{rank}(PAQ) = m + n - \text{rank}(A)$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Let $A \in \mathbb{R}^{n \times n}$ be a real square matrix such that the entry in the (i, j)-th position is given by:

$$A(i, j) = i - j$$

for all $1 \leq i, j \leq n$, where $n \geq 2$.
What is the rank of A ?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let P be a square matrix such that

$$P^2 = P$$

Which of the following statements are true?

A. $\text{Trace}(P)$ is a non-integer real number
B. $\text{Trace}(P) = \text{Rank}(P)$
C. $\text{Trace}(P)$ is an integer
D. $\text{Trace}(P)$ is a purely imaginary number

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be an $n \times n$ matrix such that the first 3 rows of A are linearly independent and the first 5 columns of A are linearly independent. Which of the following statements are true?

A. A has at least 5 linearly independent rows.
B. $3 \leq \text{rank}(A) \leq 5$
C. $\text{rank}(A) \geq 5$
D. $\text{rank}\left(A^2\right) \geq 5$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 02sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be a 5×5 matrix, and suppose that the dimension of the solution space of the homogeneous system $Ax = 0$ is at least 2 .
Then, which of the following statements are true?

A. $\text{Rank}\left(A^2\right) \leq 3$

B. $\text{Rank}\left(A^2\right) \geq 3$

C. $\text{Rank}\left(A^2\right) = 3$

D. $\det\left(A^2\right) = 0$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be a 4×5 real matrix row-equivalent to

$$R = \begin{bmatrix} 1 & -2 & -1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Let $\mathbf{c} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}, \mathbf{d} = \begin{bmatrix} y \\ 3 \\ 4 \\ 5 \\ 5 \end{bmatrix}$ be two solutions to $A\mathbf{x} = \mathbf{b}$, for some fixed $\mathbf{b} \in \mathbb{R}^4$.

Find the value of $y =$.

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be an $n \times n$ matrix and let b be an $n \times 1$ column vector. Suppose the equation

$$Ax = b$$

has no solution. Which of the following statements are true?

A. There exists a vector $c \in \mathbb{R}^n$ such that $Ax = c$ has a unique solution.

B. There exist infinitely many vectors $c \in \mathbb{R}^n$ such that $Ax = c$ has no solution.
C. If y is the first column of A, then $Ax = y$ has a unique solution.
D. $\det(A) = 0$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let

$$A = \begin{bmatrix} 2 & 0 & 5 \\ 1 & 2 & 3 \\ -1 & 5 & 1 \end{bmatrix}$$

Consider the system of linear equations $Ax = y$, where $x \in \mathbb{R}^3$ and $y \in \mathbb{R}^3$. For which vectors y does this system have a solution?
Which one of the following statements is true?

A. The system has a solution only for $y = [r \quad 0 \quad 0]^T, r \in \mathbb{R}$

B. The system has a solution only for $y = [0 \quad s \quad 0]^T, s \in \mathbb{R}$

C. The system has a solution only for $y = [0 \quad s \quad t]^T, s, t \in \mathbb{R}$

D. The system has a solution for all $y \in \mathbb{R}^3$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A and B be $n \times n$ real matrices.

- We say the determinant is additive if $\det(A + B) = \det(A) + \det(B)$
- We say the determinant is multiplicative if $\det(AB) = \det(A) \cdot \det(B)$

Which of the following statements is correct?

A. Determinant is both additive and multiplicative
B. Determinant is additive but not multiplicative
C. Determinant is multiplicative but not additive
D. Determinant is neither additive nor multiplicative

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

For $t \in \mathbb{R}$, define

$$M(t) = \begin{bmatrix} 1 & t & 0 \\ 1 & 1 & t^2 \\ 0 & 1 & 1 \end{bmatrix}$$

Which of the following statements is true?

A. $\det(M(t))$ is a polynomial of degree 3 in t

B. $\det(M(t)) = 0 \quad \forall t \in \mathbb{R}$

C. $\det(M(t)) = 0$ for infinitely many values of $t \in \mathbb{R}$

D. $\det(M(t)) = 0$ for exactly two values of $t \in \mathbb{R}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $A \in \mathbb{R}^{n \times n}$, and let $\text{adj}(A)$ denote the adjugate of A.

Which of the following statements are true?

A. If $\text{rank}(A) = n$, then $\text{rank}(\text{adj}(A)) = n$

B. If $\text{rank}(A) = n - 1$, then $\text{rank}(\text{adj}(A)) = 1$

C. If $\text{rank}(A) < n - 1$, then $\text{adj}(A) = 0$

D. If $\text{rank}(A) = n - 1$, then $\text{adj}(A) = 0$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A be an $n \times n$ real matrix that is singular (i.e., not invertible).
Let $x_0 \in \mathbb{R}^n$ and $b \in \mathbb{R}^n$ such that the equation

$$Ax_0 = b$$

is satisfied.
Which of the following statements are true?

A. There exists a vector $y_0 \in \mathbb{R}^n$ such that $A^T y_0 = b$.
B. There exist infinitely many solutions to the equation $Ax = b$.
C. If $A^T x = 0$, then $b^T x = 0$.
D. The solution to the equation $Ax = b$ is unique.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A be an $m \times m$ real matrix, and let $x \in \mathbb{R}^m$ be a column vector of unknowns. Consider the following two statements:

Statement 1: There exists a non-zero vector $b_1 \in \mathbb{R}^m$ such that the linear system $Ax = b_1$ has no solution.
Statement 2: There exist non-zero vectors $b_2, b_3 \in \mathbb{R}^m$, with $b_2 \neq cb_3$ for any scalar $c \in \mathbb{R}$, such that both

systems $Ax = b_2$ and $Ax = b_3$ have solutions.

Which of the following options is/are correct?

A. Statement 2 is true whenever A is singular.
B. Statement 1 is true whenever A is singular.
C. Both Statement 1 and Statement 2 can be true simultaneously.
D. If $m = 2$, then at least one of Statement 1 and Statement 2 is false.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A be a 2×2 real matrix such that

$$\det(A) = 1 \quad \text{and} \quad \operatorname{tr}(A) = 3.$$

What is the value of $\operatorname{tr}\left(A^2\right)$?

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

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